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Structuring Data for Analytics

Learning Session

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Agenda

Structuring Data for Analytics

- A Little About Me
- Understanding Source System Data
- Structuring Data for Analytics
  - Identify the Macro Entities
  - Simplify the Query Paths
  - Add Classification Hierarchies
  - Flatten the Data
  - Integrate the Facts

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A Little About Me

- Built Data Warehouses for:
  - Wells Fargo
  - American Electric Power
  - Carlyle Group
  - Barnes and Noble
  - JM Family (Toyota of South East U.S.)
  - U.S. State Department
  - Century Link
  - …

Work History:

AMS (now CGI), PricewaterhouseCoopers, Business Objects, SAS (twice)

Source System Data

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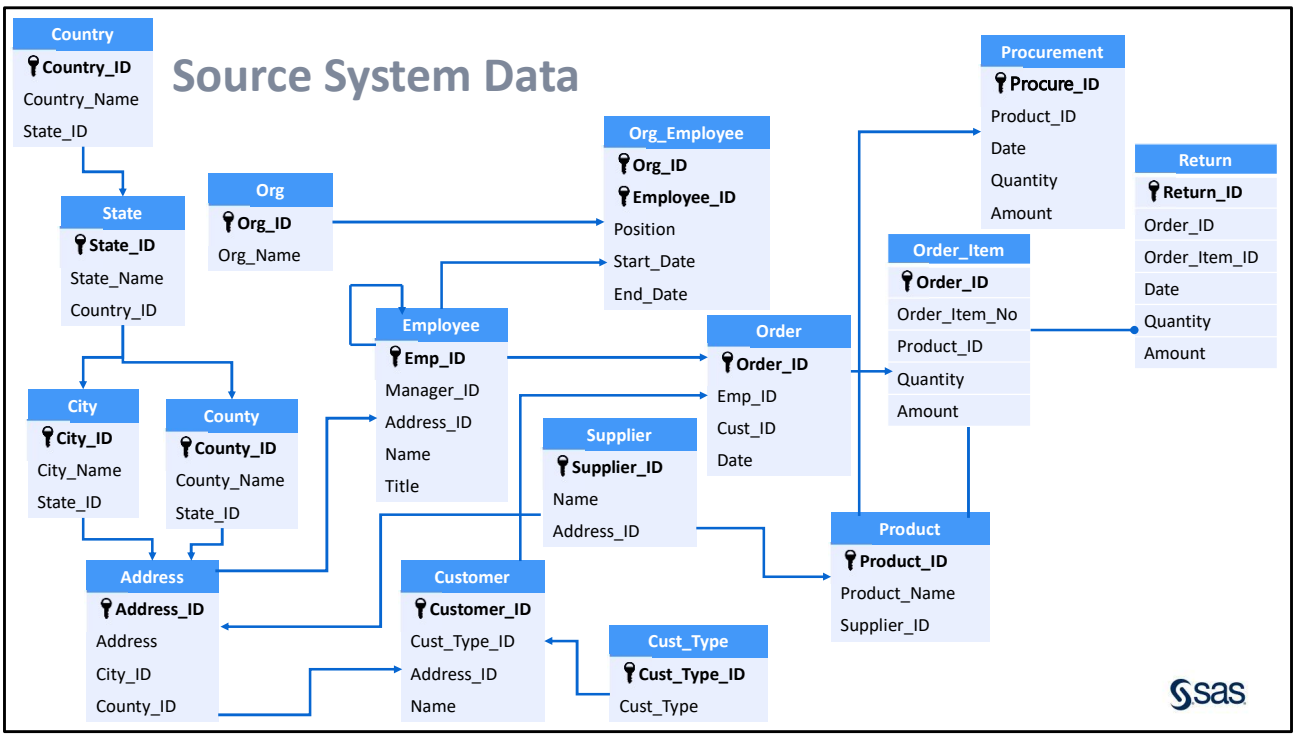

True source data is almost always transactional in nature. Some exceptions like survey data do exist.

Transactional data is organized to eliminate data inconsistency by removing data redundancy. Therefore, each entity like organization, geographic region, etc. will have its own table instead of repeating on its macro entity (employee, geography, order, …)

This is the opposite of how we usually like the data organized for analytics where we like all the data combined.

Operational system data has too many join paths, too many *possible* join paths, and multiple data grains.

How do we unpack this maze of tables?

Identify Macro Entities

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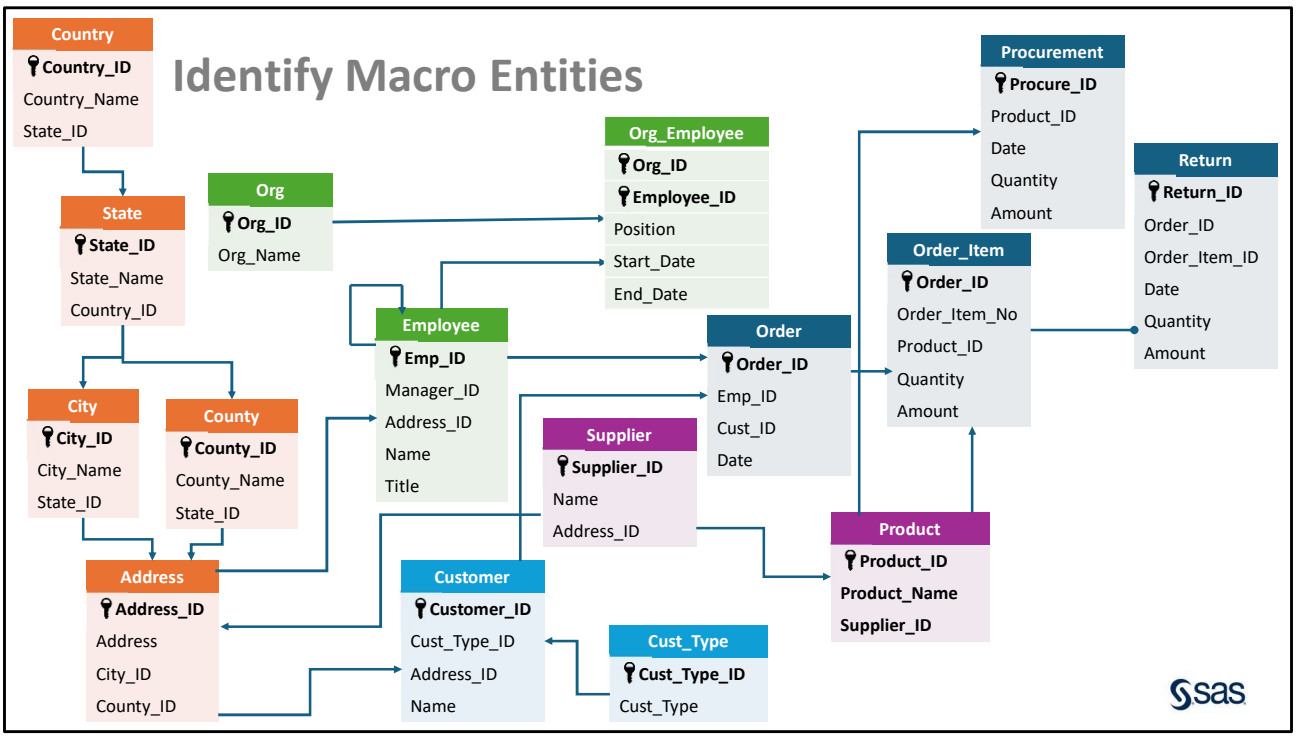
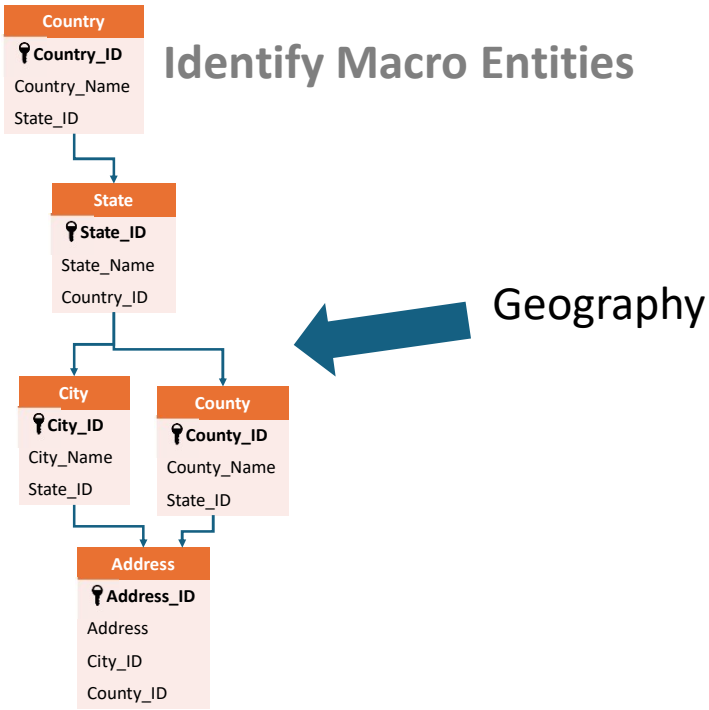
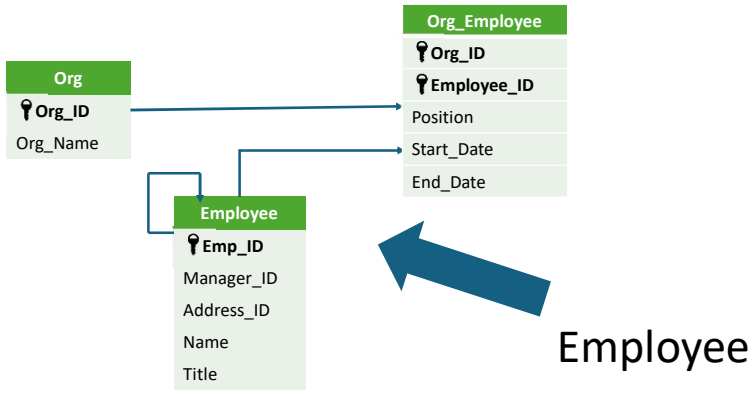


First, we group the tables into macro entities.
For retail data, entities like customer, supplier, employee, geography, as well as the transactional entities like order, procurement, returns, inventory, etc.

Identify Macro Entities

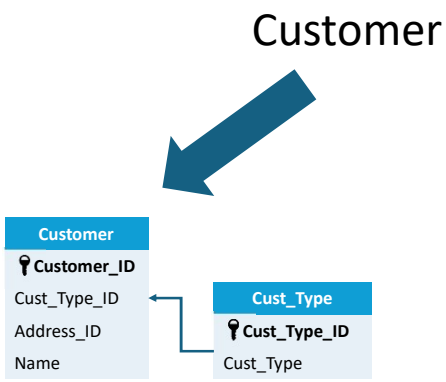


Identify Macro Entities

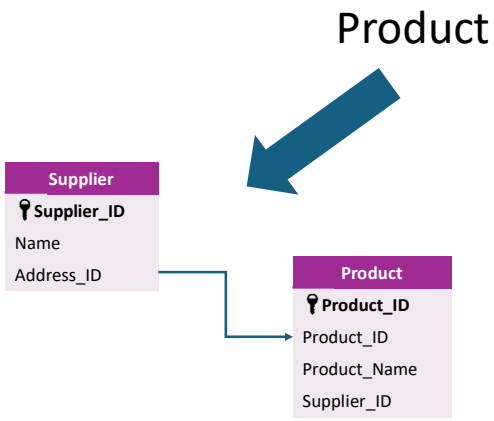


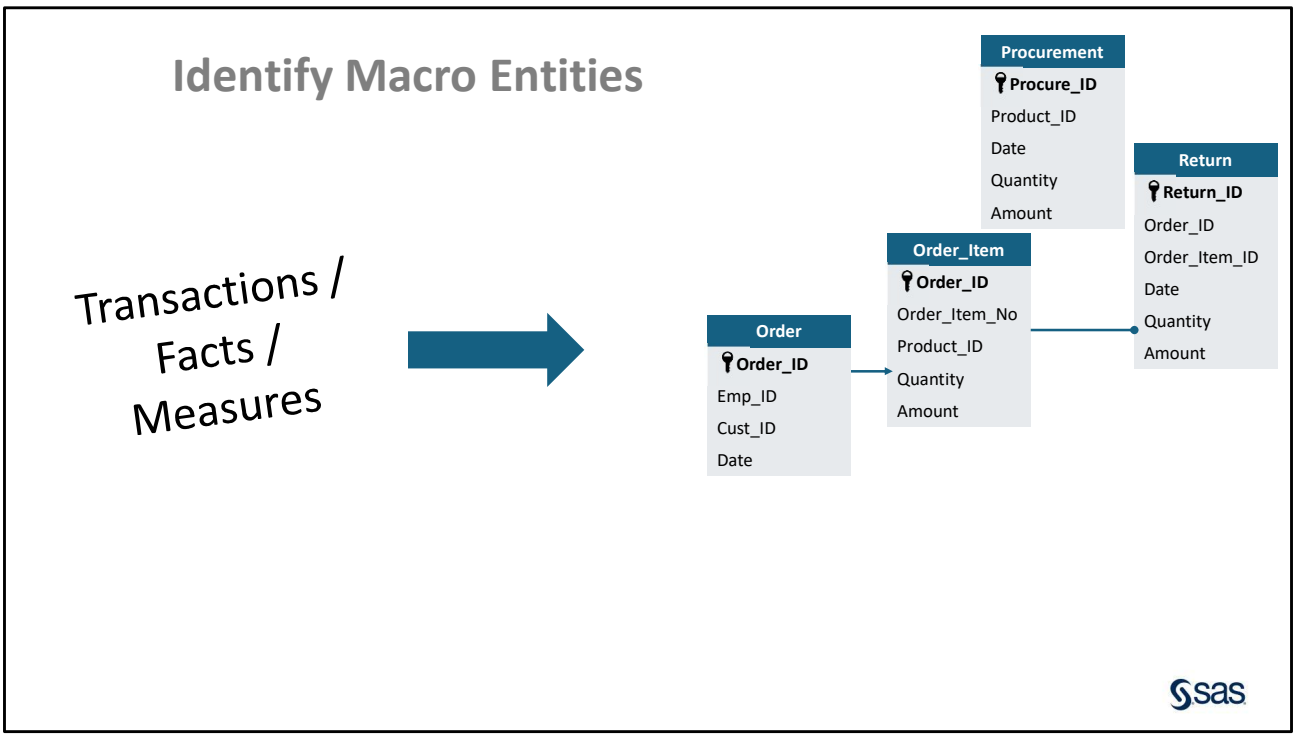
Employee

Identify Macro Entities



Identify Macro Entities





Order, Return, Procurement
In most sample data, you'll see measure data like this side by side on the same record.
For example, cost will appear right next to order amount.
This is not realistic.
Data coming from operational systems will represent the actual transactions that happened.
Retail products will generally be bought in bulk and represented in a set of procurement tables.
To model this data for analytics, we will want to keep this basic structure but enhance it to combine these various transaction types.

Simplify Query Paths

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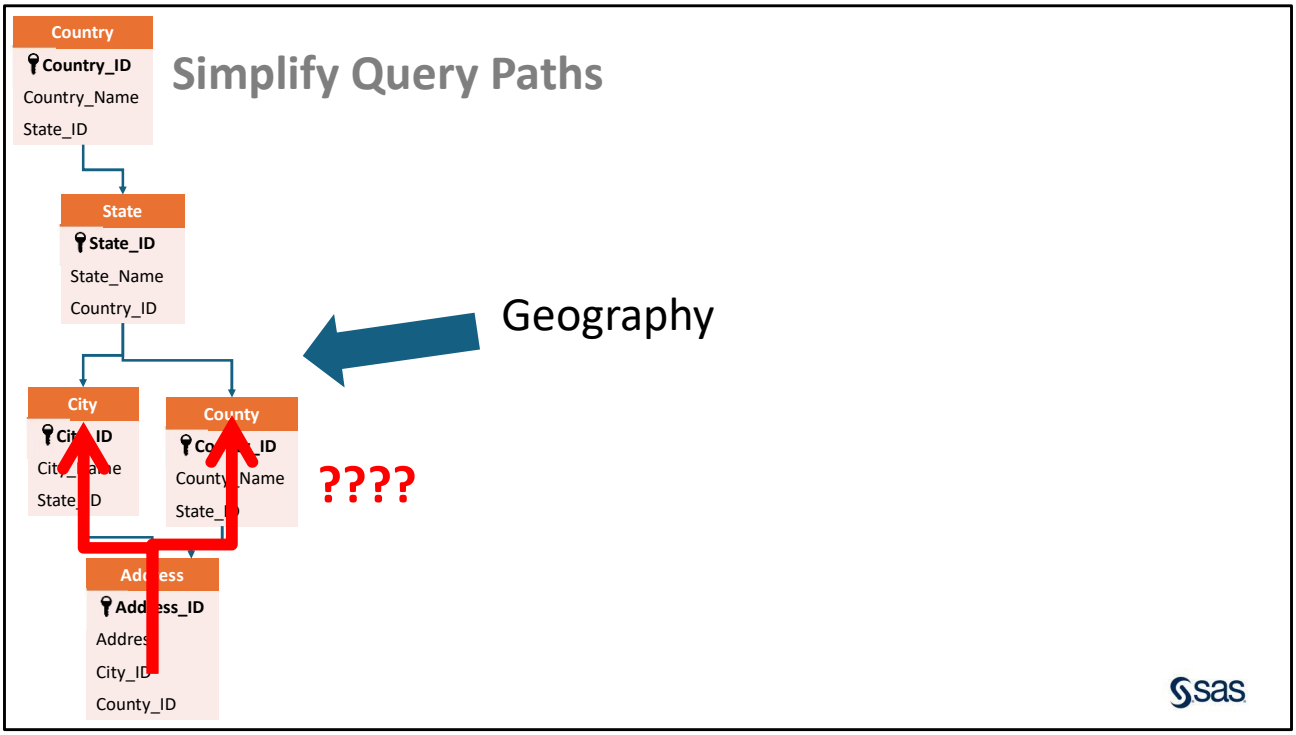
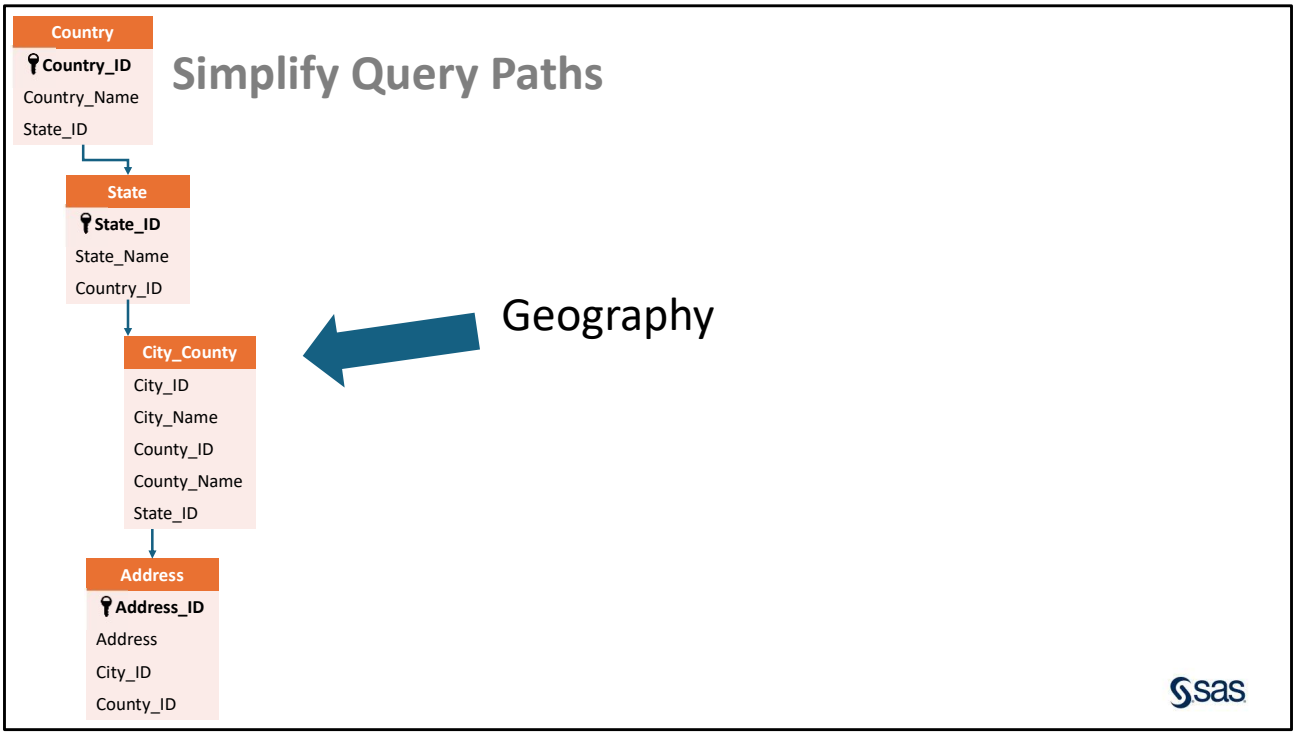
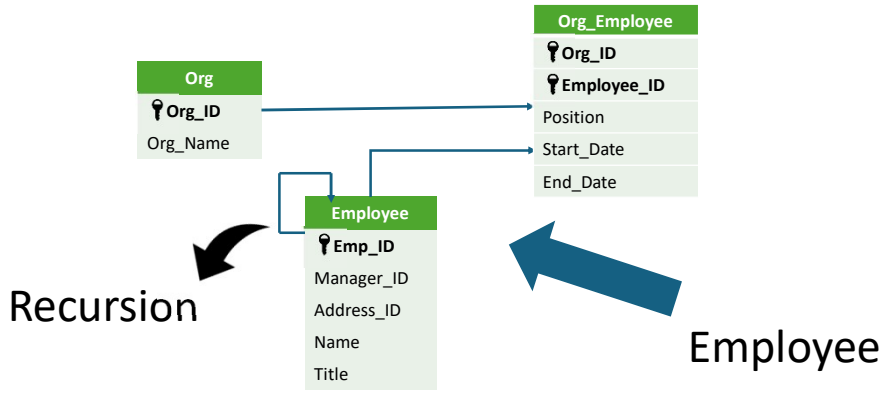


Once our macro entities are determined, we need to determine its correct query path and remove any ambiguity in how the tables will be joined.
Some US states like Virginia have independent cities which exist separate from counties.
Other states have cities within counties. So, addresses would have both.
An end-user or query tool may not choose the correct join path.



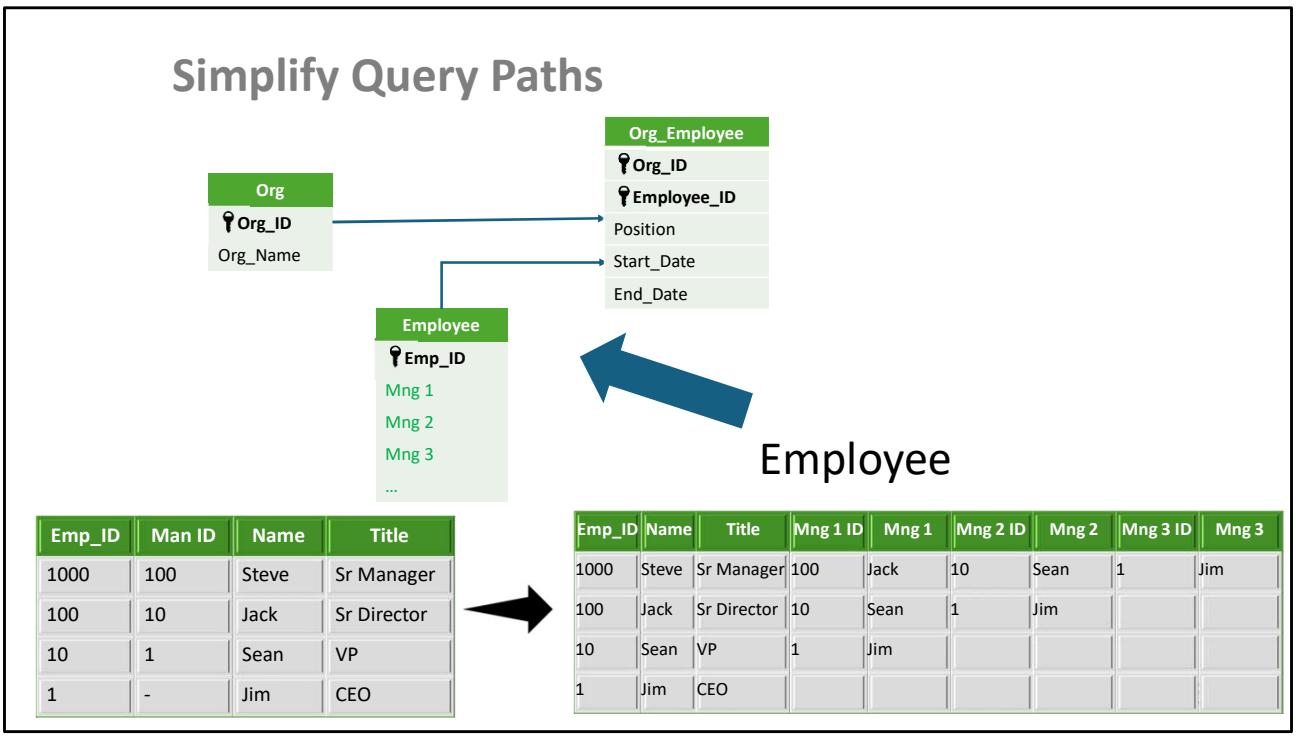
Here we combine the two entities to ensure each address is built correctly.

Simplify Query Paths



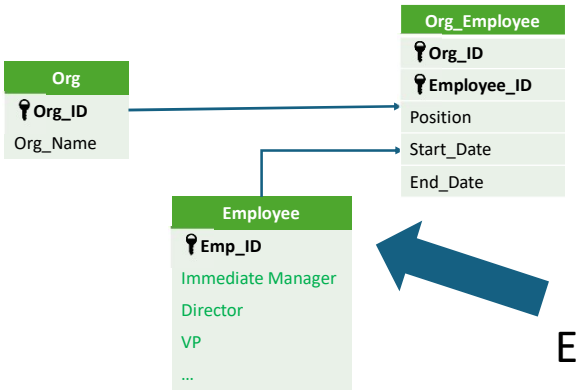
Emp_ID	Manager_ID	Name	Title
1000	100	Steve	Sr Manager
100	10	Jack	Sr Director
10	1	Sean	VP
1	-	Jim	CEO

The employee table here uses recursion.
Records reference other records in the same table to represent employee-manager relationships.
We need to unspin this to make it useful for analytics.



We could simply unspin the table into a wider table that represents all the relationships for each employee.
However, this isn't super useful either as Manager 1 for one employee may be a mid-level manager but a VP for another.

Simplify Query Paths

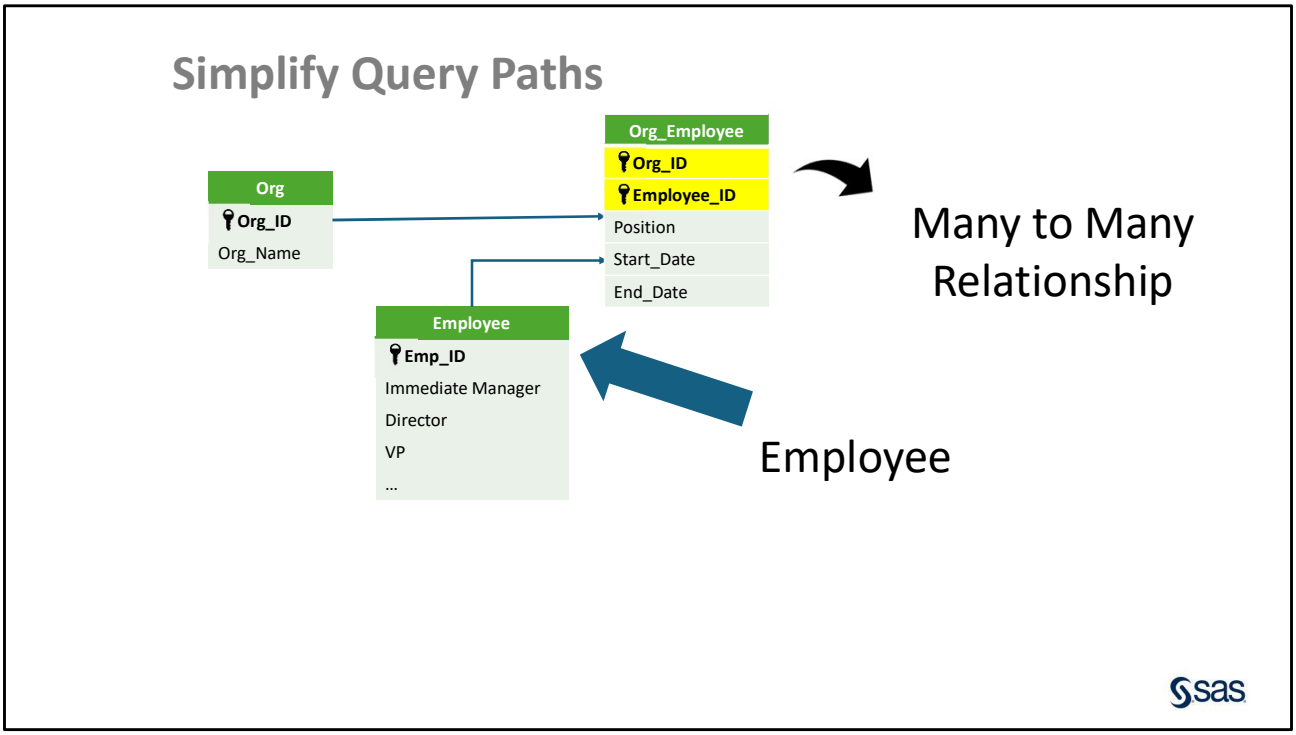


Employee

Emp_ID	Man ID	Name	Title
1000	100	Steve	Sr Manager
100	10	Jack	Sr Director
10	1	Sean	VP
1	-	Jim	CEO

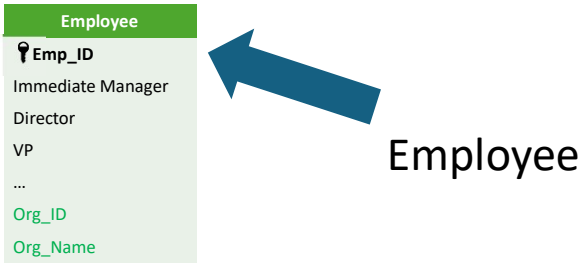
Emp_ID	Name	Title	Immediate Mng 1 ID	Immediate Mng 1	VP ID	VP Name
1000	Steve	Sr Manager	100	Jack	10	Sean
100	Jack	Sr Director	10	Sean	10	Sean
10	Sean	VP	1	Jim	10	Sean
1	Jim	CEO				

A better way is to pick common levels like director, division chief, VP, etc. that represent the business well.



Here we have a many-to-many relationship between employee and org.
Over time, employees belong to different orgs as they take different positions within the company.
This is useful information but generally too hard to unpack at query time.
We want to make queries easy for our users and minimize mistakes.

Simplify Query Paths



For analytics, we'll have to pick a single org and associate it with the employee record.
This may be the current org.
It could also be the org at the time of the transaction.
This would mean employees would have multiple records that join to the transactions from the time period he/she belonged to the org. (Slowly changing dimension)

Add Classification Hierarchies

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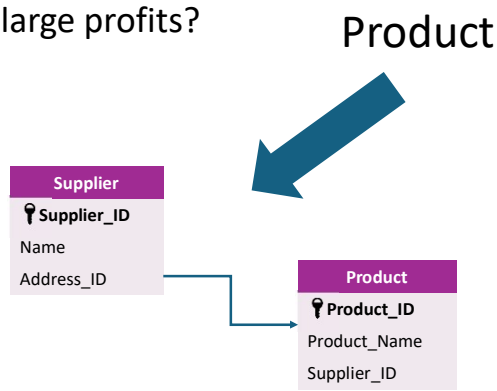
Add Reporting Hierarchies

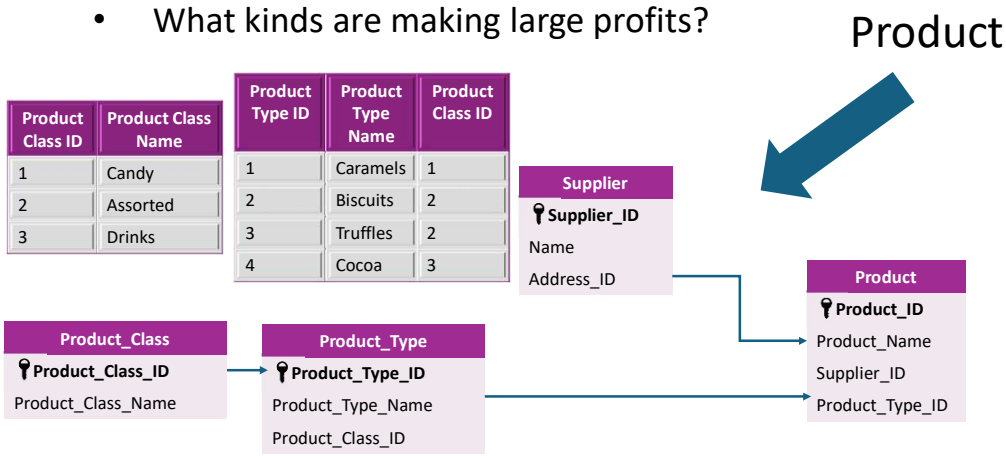
- What kinds of products are selling?
- What kinds are being returned?
- What kinds are making large profits?



Our product data is very thin. All we have is product name and supplier.
This is fine for basic transactional reporting but for analytics we need more levels.
In general, we want to create many different hierarchies with multiple levels inside.
This gives us many ways to group the data and answer questions like those above.

Add Reporting Hierarchies

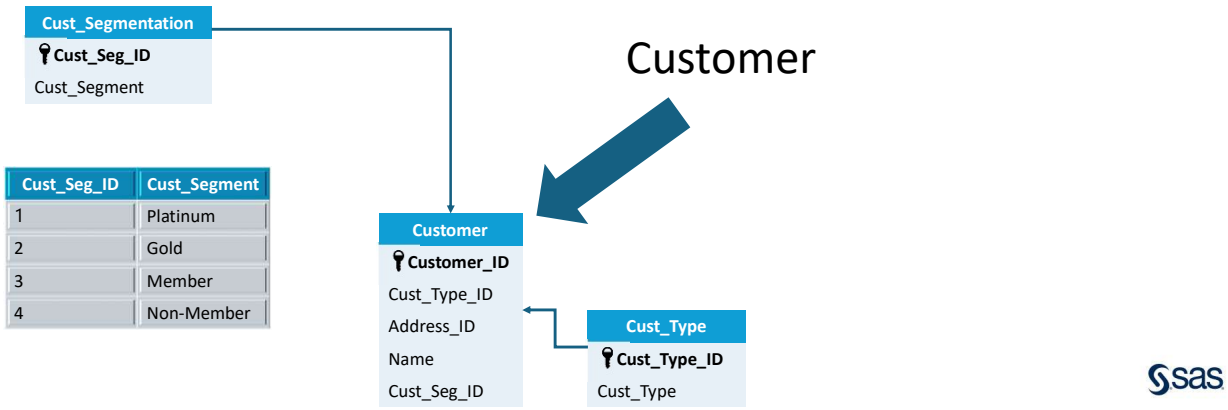
- What kinds of products are selling?
- What kinds are being returned?
- What kinds are making large profits?



Work with the domain owners to create product rollups that create context for analytics. In the example, we add product type and product class that add richness to the data. With these we can see the categories of product that are selling or are most being returned, etc.

Add Reporting Hierarchies

- Demographics
- Multiple Segmentation Models
- Lifetime Value
- …



Cust_Seg_ID	Cust_Segment
1	Platinum
2	Gold
3	Member
4	Non-Member

Again, our customer data is very thin.
All we have is customer type which might represent whether customers are in-person, on-line, corporate accounts, etc.
Adding demographics, or segmentation scores from the marketing group will make the data much more useful.
Here we add a single segmentation score that groups our customers into 4 value groups.
You can add multiple segmentation scores, data from call centers, data from marketing campaigns, and other touchpoints to add richness.

Flatten the Data

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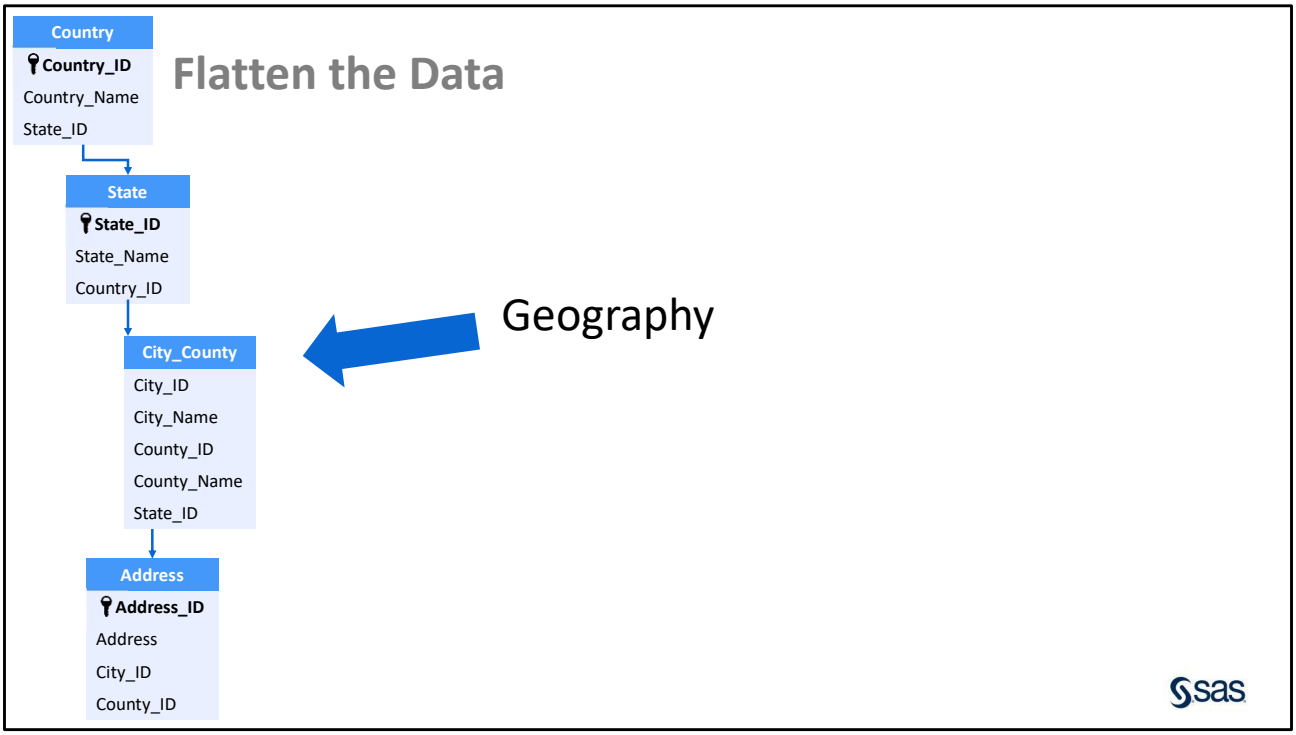


Depending on your analytics tool of choice, this step is not required.
You wouldn't need to do it with SAS Web Report Studio for example as the information map would do it for you.
However, for Visual Analytics, it is required.
VA likes flat data.
It can work with fully flattened tables as well as CAS star-schema views.
Flattening can be down with any number of approaches including data pipelines (i.e. traditional ETL), database or SAS views, semantic layer tools, VA data views, VA data joins, etc..

Flatten the Data

Geography
Geo_ID
Address_ID
City_ID
County_ID
State_ID
Country_ID
Address
City_Name
County_Name
State_Name
Country_Name

Geography

Flatten the Data

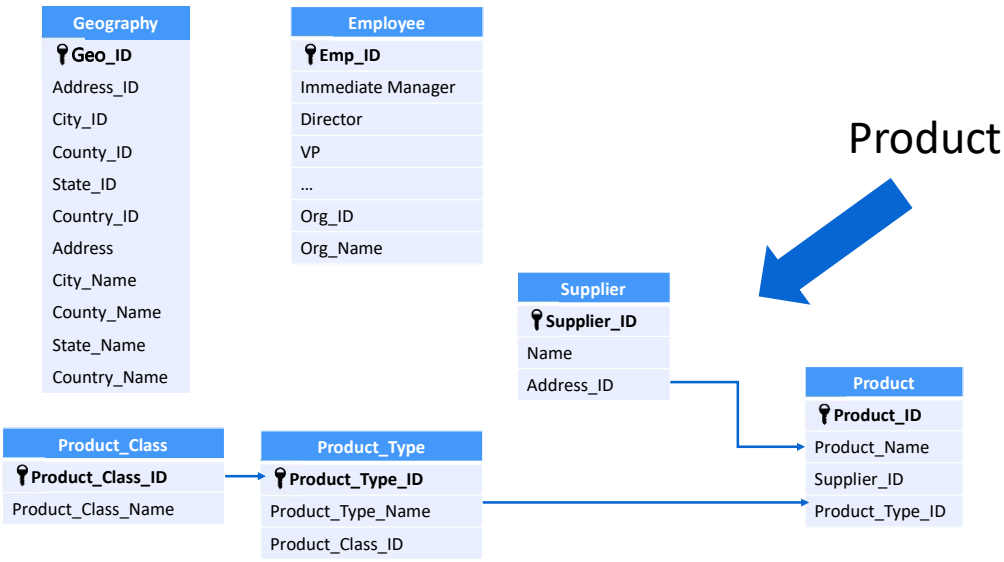
Geography
Geo_ID
Address_ID
City_ID
County_ID
State_ID
Country_ID
Address
City_Name
County_Name
State_Name
Country_Name

Employee
Emp_ID
Immediate Manager
Director
VP
…
Org_ID
Org_Name

Employee

Employee was already flattened in our query path exercise.

Flatten the Data



Flatten the Data

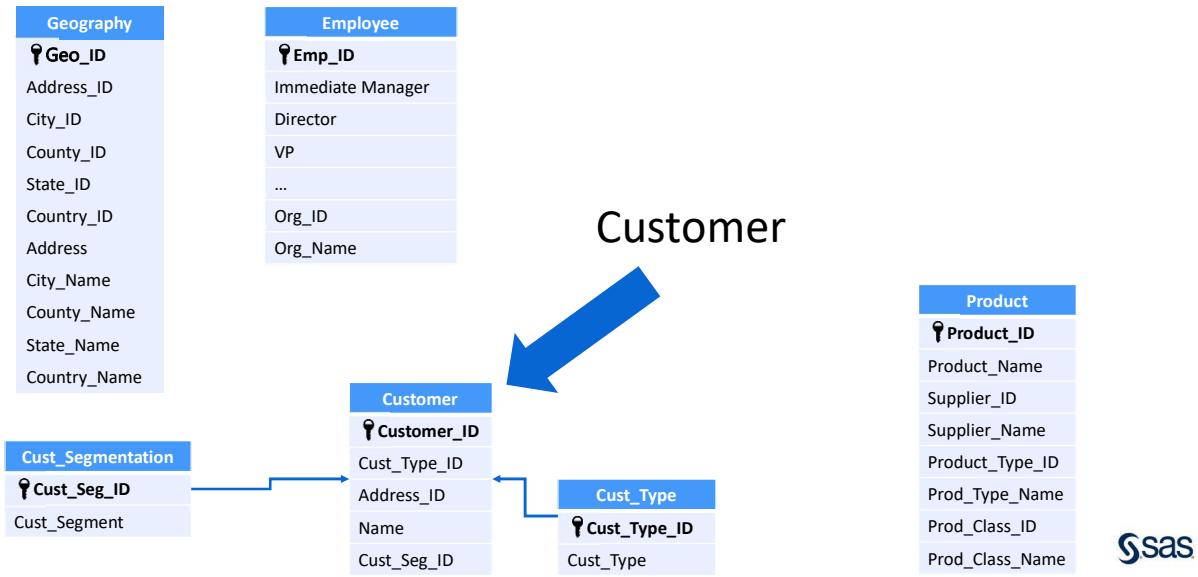
Geography
Geo_ID
Address_ID
City_ID
County_ID
State_ID
Country_ID
Address
City_Name
County_Name
State_Name
Country_Name

Employee
Emp_ID
Immediate Manager
Director
VP
…
Org_ID
Org_Name

Product

Product
Product_ID
Product_Name
Supplier_ID
Supplier_Name
Product_Type_ID
Prod_Type_Name
Prod_Class_ID
Prod_Class_Name

Flatten the Data



Flatten the Data

Geography
Geo_ID
Address_ID
City_ID
County_ID
State_ID
Country_ID
Address
City_Name
County_Name
State_Name
Country_Name

Employee
Emp_ID
Immediate Manager
Director
VP
…
Org_ID
Org_Name

Customer

Customer
Customer_ID
Cust_Type_ID
Cust_Type
Name
Cust_Seg_ID
Cust_Segment

Product
Product_ID
Product_Name
Supplier_ID
Supplier_Name
Product_Type_ID
Prod_Type_Name
Prod_Class_ID
Prod_Class_Name

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Integrate the Measures

Which method works best?

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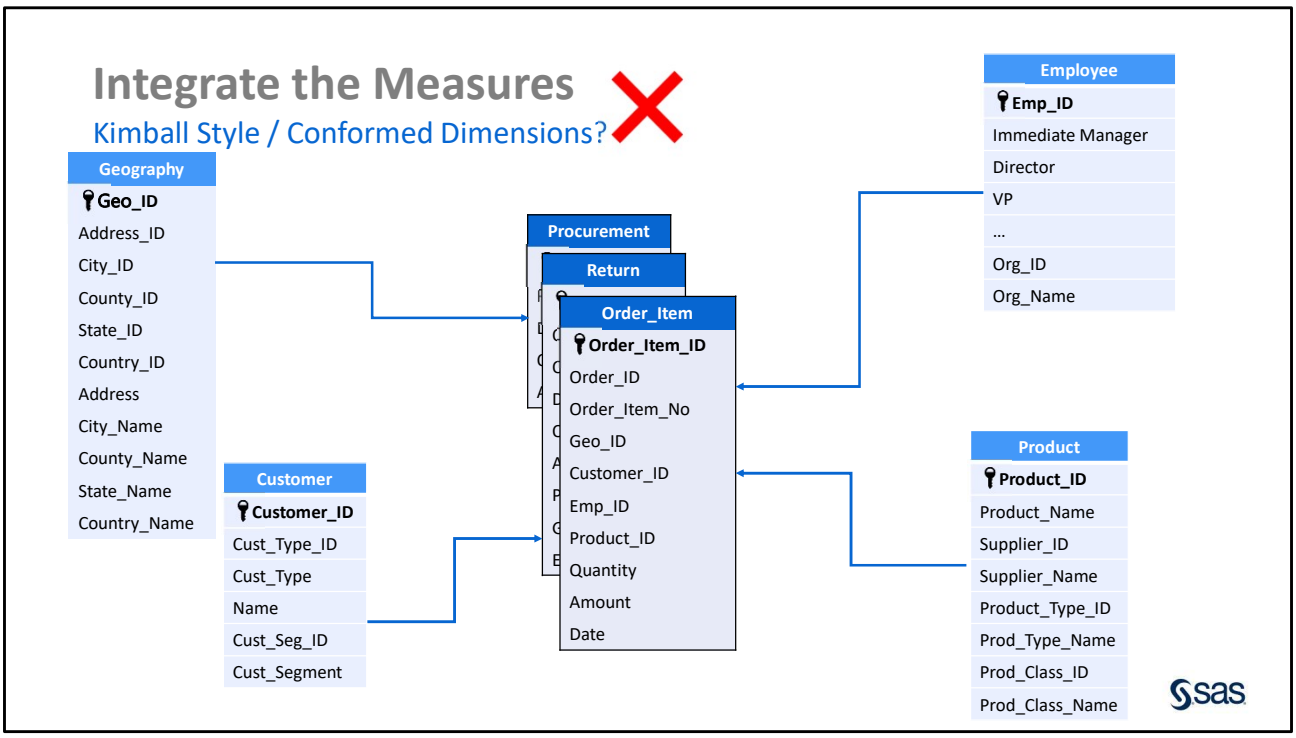


Ralph Kimball touted a design style that measures be kept on separate “Fact Tables” which each join in the same way to a set of “conformed dimension” tables, i.e. our flattened entity tables.

So, one table for returns, one for order, one for procurement, etc.

This approach is fine and probably the easiest to achieve with views and minimal ETL.

However, we won’t recommend it for getting the most analytic value out of your data.

Integrate the Measures

Kimball Style / Conformed Dimensions?

order_item_id	order_id	or...	Emp_ID	Geo_ID	Customer_ID	Product_ID	date	Quantity	Amount
2697	000000001581	1	1000001021	S1000041	4967020	1030	18831	30	13.29529703
4981	000000001537	2	1000003021	S1000041	60054020	1030	17857	28	12.408943894

return_id	order_id	Geo_ID	Customer_ID	Product_ID	Emp_ID	date	Quantity	Amount
1160	000000001581	S1000041	4967020	1030	1000001021	18841	30	13.29529703
8819	000000001657	S1000041	7933020	3030	1000005021	17570	8	3.5454125413

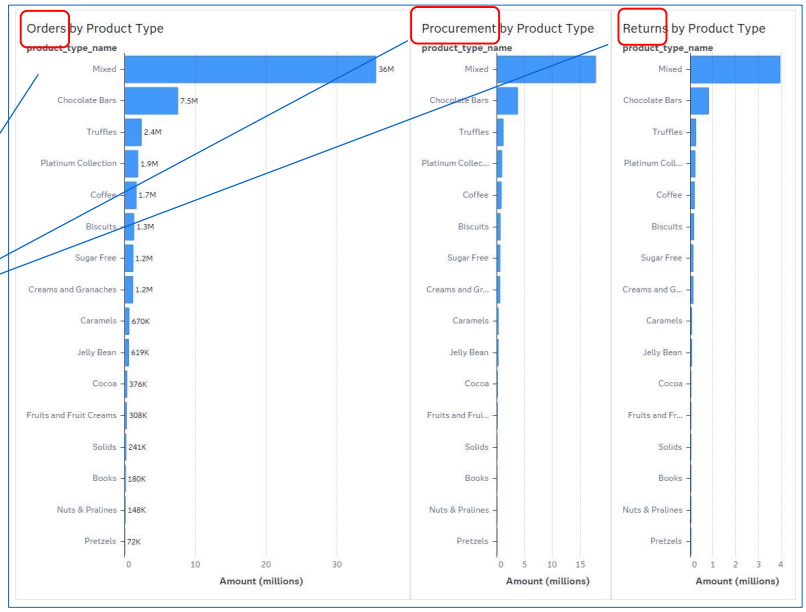
procure_id	Product_ID	Emp_ID	date	Quantity	Amount	org_name	product_type_name
2163	1030	1000001021	18631	30	6.6476485149	Sales	Biscuits
2093	1030	1000003021	17657	28	6.2044719472	Sales	Biscuits

Here is a subset of the data fully combined.
There is one table/view for order, another for returns, and another for procurement.
Each has the same categorical fields where appropriate. These are the conformed dimensions.

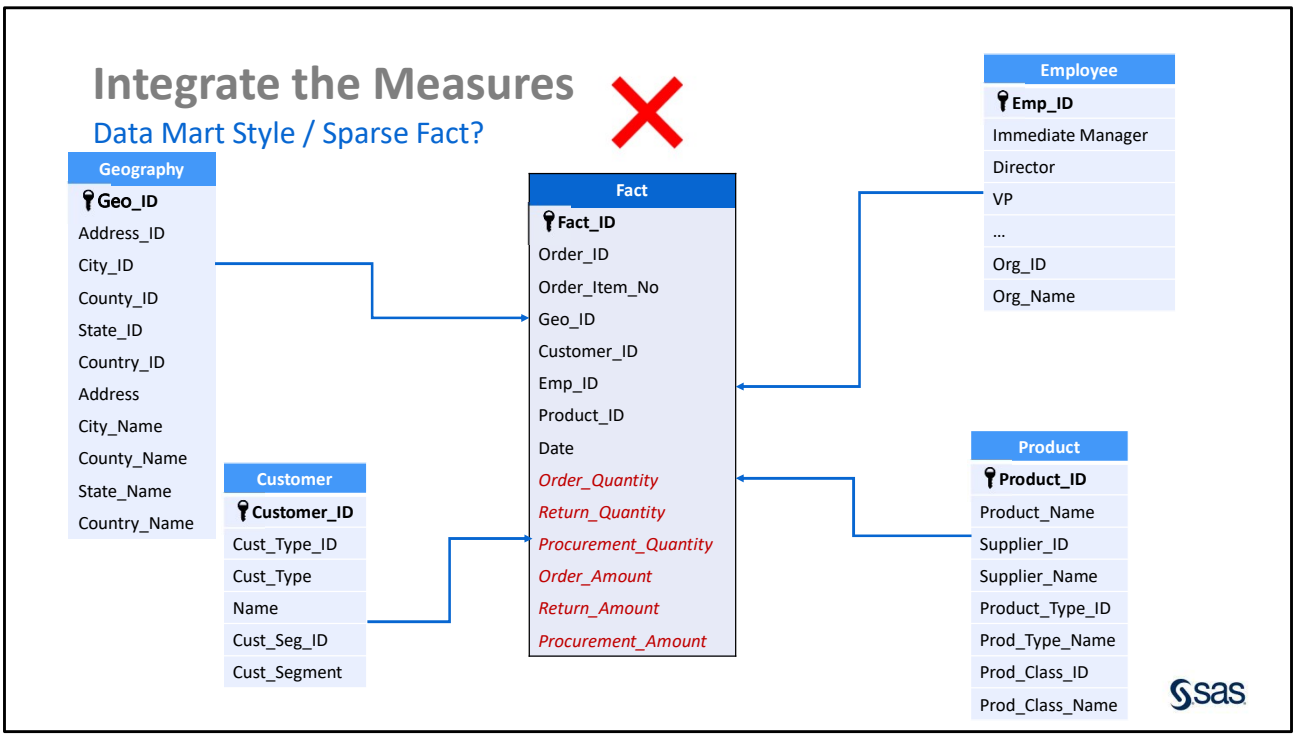
Integrate the Measures

Conformed Dimensions

Measure Integration is Limited



Using separate fact tables, the closest we can get to integrating measures in the same analysis is putting them side-by-side in a report.
This is useful no doubt, but we can do better.
We can of course use the “Join” feature in Visual Analytics, but this will just take us to the other approaches we show next.



How about combining the measure data onto a single central table?
So now instead of separate returns, orders, and procurement tables, we have a generic central table that combines the measures from each.

Integrate the Measures

Data Mart Style / Sparse Fact?

t_segment	org_name	product_type_name	date	order_quantity	order_amount	return_quantity	return_amount	procurement_quantity	procurement_amount
late Club Member	Sales	Creams and Granaches	18405	37	18.214257426	.	.	.	.
Member	Sales	Creams and Granaches	18586	.	.	1	0.4744018152	.	.
	Sales	Caramels	18209	.	.	.	.	7	1.75

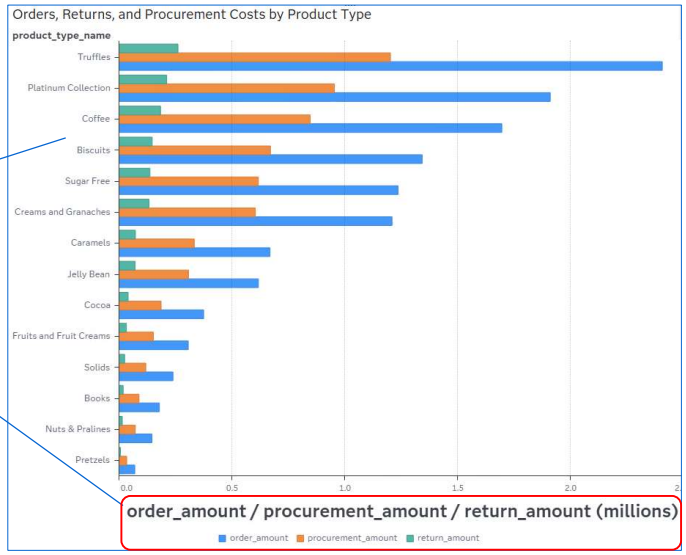
Here is a subset of the data fully combined.
The measure fields are strung out like in a classical data mart.
If the row is an order row, the other two sets of measures are null.
If the row is a return row, the other two sets of measures are null….
Thus, it is sparse.

Integrate the Facts

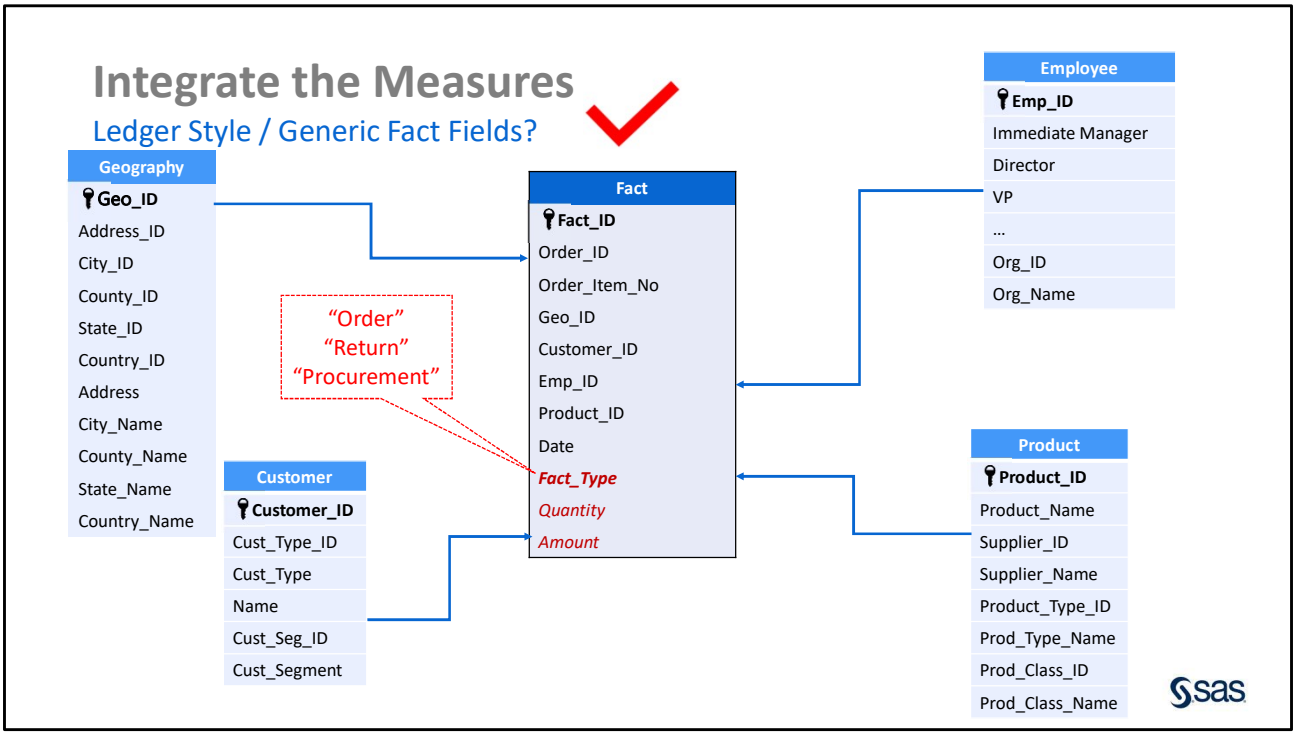
Sparse Fact

Facts can be Added, Compared, …

if the VA object allows more than one measure



This approach works better since we can directly integrate different measures within VA objects that allow more than one measure.
In the example, we can directly compare order amounts with return and procurement cost amounts on the bar chart object.



However, we can do even better by using generic measure fields, amount and quantity in the example, and adding a "Fact Type" field.
This would be called a degenerate dimension in Kimball terminology.
The fact type determines the type of row represented. "Order" for an order row. "Return" for a return row…
Again, each row type only joins to the dimension tables appropriate to it.

Integrate the Measures

Ledger Style / Generic Fact Fields?

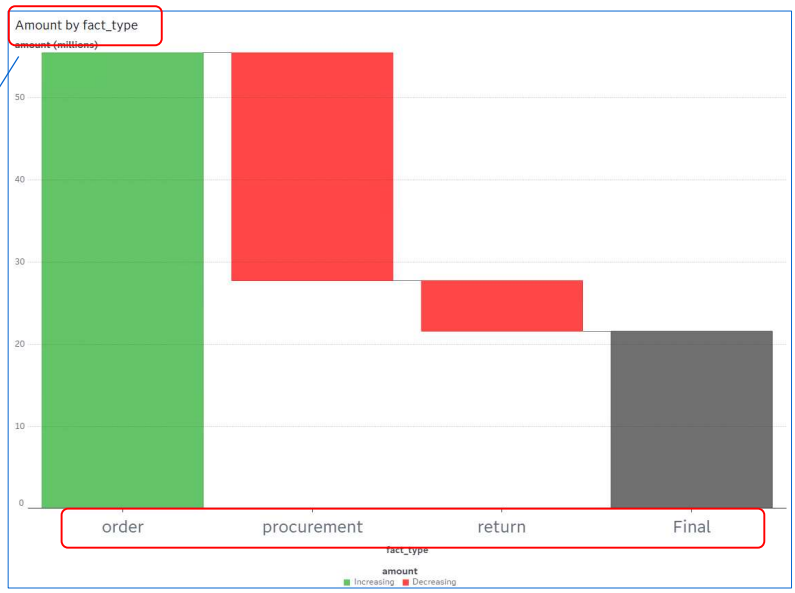
geo_id	customer_id	Product_ID	state_name	cust_segment	product_type_name	date	fact_type	Quantity	amount
	.	34030			Creams and Granaches	18205	procurement	37	-9.107128713
S1057041	1	34030	OH	Non-Member	Creams and Granaches	18586	return	1	-0.474401815
S8000041	36047020	20030	NC	Chocolate Club Member	Caramels	18409	order	7	3.5

Here is a subset of the data fully combined.
The measure fields are left with the generic field names, quantity and amount.
The fact type field denote the type of record.
Note that we made the amounts negative for procurement as well as return records since these denotes costs / lost revenue.
This can be done in the data or at report time.

Integrate the Facts

Ledger Style

Separate Facts used within a single VA Object



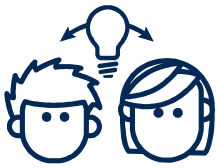
Modeling the data like this we can directly use multiple measures in all VA objects, even ones that only allow one measure.
Here we see the waterfall chart which only tracks one measure.
Plotting amounts by fact type, we can see how much of the order revenue is taken up by returns and procurement costs.

Integrate the Facts

Ledger Style

Plus, you can define separate measure fields as well

This approach also gives you the best of both worlds since you can define calculated items for the separate measure fields as well.
So, you can have a ledger style table as well as a sparse fact table all in one.



Structuring Data for Analytics

- Identify the Data's Macro Entities
- Simplify the Query Paths
- Add Classification Hierarchies
- Flatten the Data
- Integrate the Facts

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